

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA
Course: Software Engineering

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Software engineering. Highlight the key differences between Software Engineering and System Engineering. Also, describe any five attributes of good software with suitable examples. 7
- b) What is software specification? Explain how UML diagrams such as Use Case diagram, State Diagram and Activity diagram can be used to represent software behavior. 8
2. a) What is software testing? Describe any five levels of testing with appropriate examples. 7
- b) What are test case? Explain the major differences between Black Box Testing and White Box Testing with examples. 8
3. a) What are defect amplification and defect removal, and how does the review method help reduce errors? Please explain with a diagram. 7
- b) What is the relationship between reliability, availability, and serviceability in software systems, and how are these factors interdependent? Discuss in detail. 8
4. a) What is a project estimation technique? Explain the Basic COCOMO model in detail. 7
- b) What are the current key trends in software engineering, and how significant is SaaS as a major trend in today's IT industry? Discuss in detail. 8
5. a) Explain OOP concept in Project management. 8
- b) What are the differences between Alpha and Beta testing, and what are the advantages and disadvantages of each? 7
6. a) What do you understand by proof of correctness in Cleanroom Software Engineering? How does it differ from traditional defect detection techniques? 7

- b) Why do we need software configuration management? Explain its activities. 8
7. Write short notes on: (**Any two**) 2×5
- a) Attribute of Good Software
 - b) Technical review
 - c) CORBA